

B2 G4

EMPEL Lab Report

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Problem Statement (DC-DC Cuk Converter)

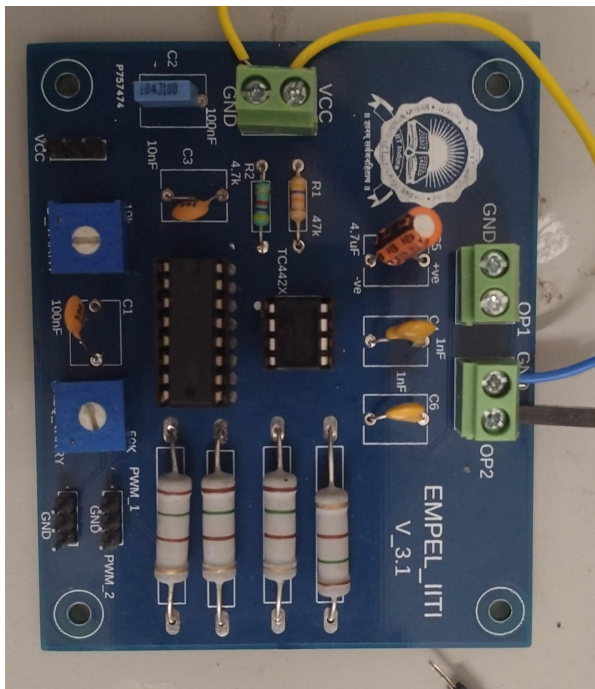
- Input: 10 V
- Frequency: 10 kHz

Output: 7.5–12.5 V
Output Current: 1 A

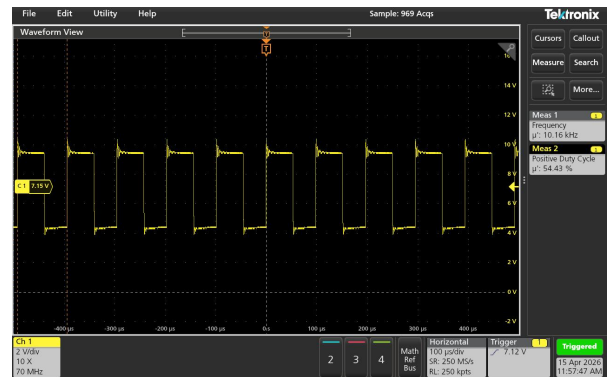
Tasks

1. Capture CCM waveforms.
2. Capture DCM waveforms via increased load resistance.

Hardware



(a) Soldered PCB



(b) Gate Driver output

Project Video Demo:

[Click here to watch the Hardware Demonstration on Google Drive](#)

Hardware Waveforms

CCM Waveforms



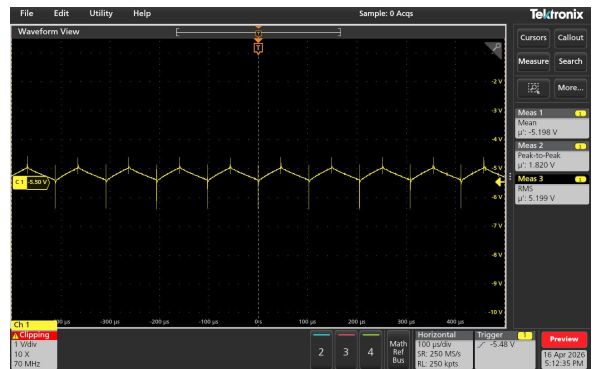
(a) L1 Voltage (CCM)



(b) L2 Voltage (CCM)



(c) Diode Current (CCM)

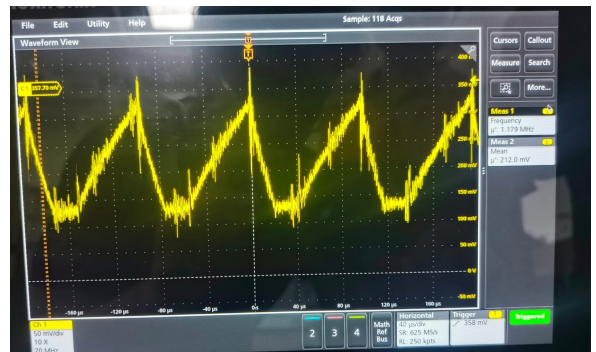


(d) Output Voltage (CCM)

DCM Waveforms



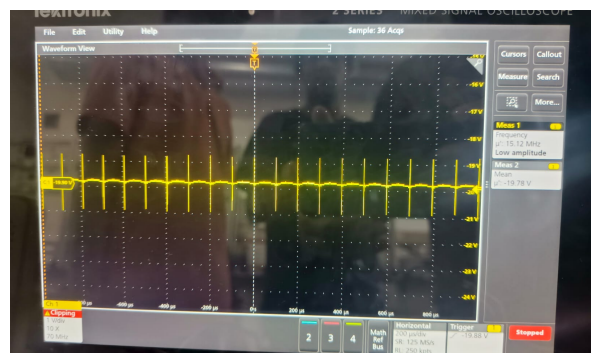
(a) L1 Voltage (DCM)



(b) L2 Voltage (DCM)



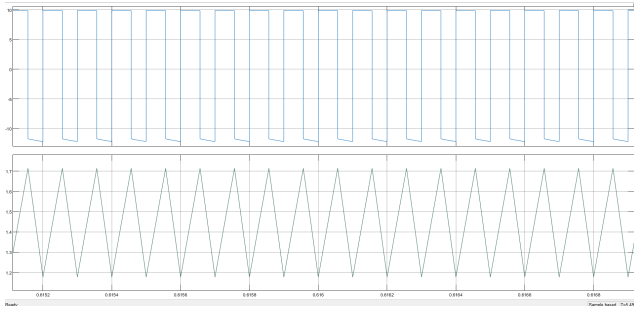
(c) Diode Current (DCM)



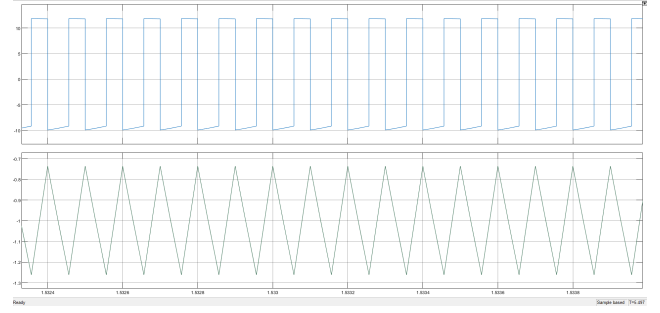
(d) Output Voltage (DCM)

Simulation Results

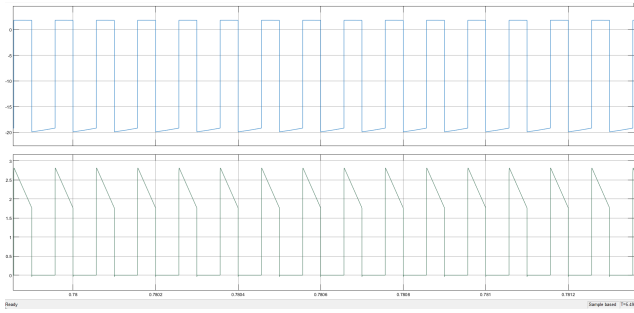
CCM Simulation



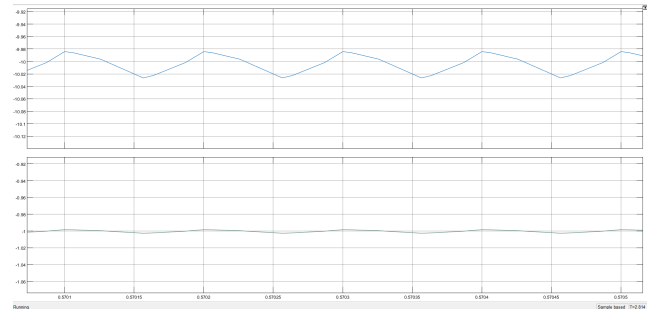
(a) Inductor 1 Current



(b) Inductor 2 Current

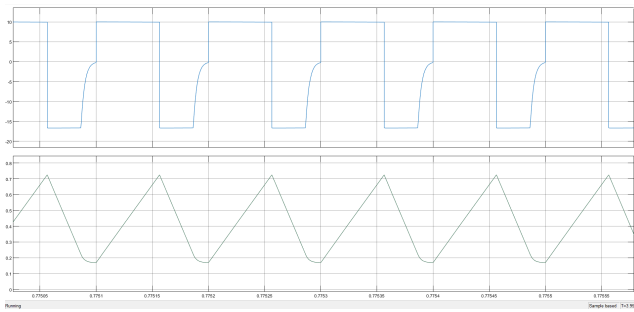


(c) Diode Current

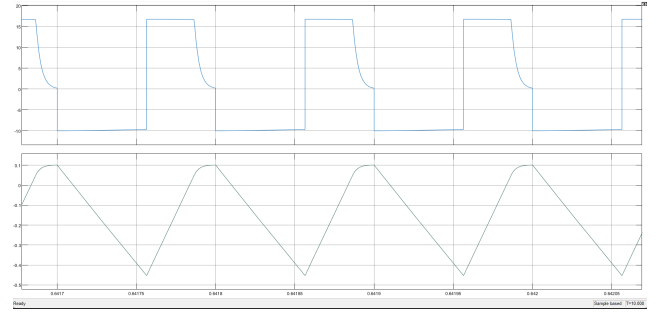


(d) Vout and Iout

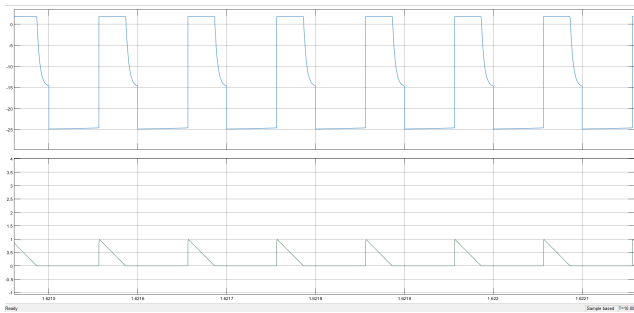
DCM Simulation



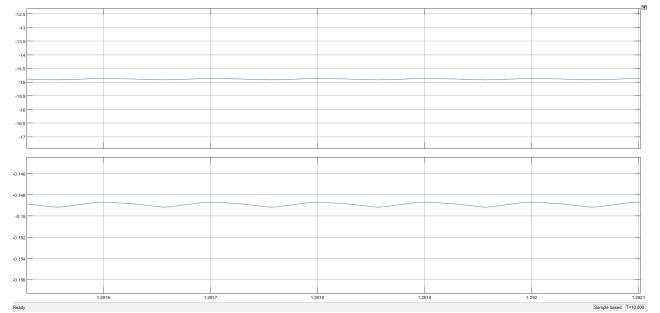
(a) Inductor 1 Current (DCM)



(b) Inductor 2 Current (DCM)



(c) Diode Current (DCM)



(d) Vout and Iout (DCM)